

Was Pierre Marie right? Not really

Stefano F. Cappa & Eleonora Catricalà

IUSS PAVIA & IRCCS MONDINO FOUNDATION, PAVIA

Pierre Marie, the arch-enemy of Jules Déjerine and Augusta Klumpke-Déjerine, is apparently enjoying a new wave of popularity (Coutinho, Caramelli & Teive 2021, Weiller, Glauche & Rijntjes 2022). While some of his anatomical work may be justly appreciated, a careful reading of his work on aphasia indicates that his theoretical stance was deeply flawed. In this commentary, we briefly discuss why, to us, Pierre Marie was, and remains, definitely not right.

The conception of aphasia in Pierre Marie

The main source on which this commentary is written consists of the first of the three papers Marie published in the journal *Semaine Médicale* between May 1905 and November 1906 (Marie 1906a,b,c). At the time Pierre Marie was working at Hospice de Bicêtre, before his appointment to the chair of Anatomie Pathologique in 1907. The two later papers are mainly dealing with anatomical issues, while the first one is largely dedicated to Marie's neuropsychological conception of aphasia. The paper starts with a strong attack against the "current aphasia doctrine", which he claims is "theoretical and schematic". As an (unfortunate) example, Marie is focusing on Wernicke's "radically false" idea of an audition centre in the first temporal gyrus, based on "Meynert's description". The reason for this attack to the

Wernicke-Lichtheim model in a paper whose title is “the left third frontal gyrus does not play any special role in language function” becomes soon clear, and is related to the central question of what aphasia is. In Marie’s words “in aphasics there is something much more important and serious than loss of word meaning; there is a *very marked reduction of intellectual capacity* in general”. This statement, to be further elaborated, indicates that Marie is following the interpretation of aphasia originally proposed by Finkelnburg (Finkelnburg 1870) as a disorder of symbolic thought, extending to multiple extra-verbal abilities (an English translation can be found in (Duffy & Liles 1979)). Marie’s position is however somewhat more precise: his claim is that the disorder is mainly affecting “the stock of things learnt by didactic procedures”, while “affective reactions” and “moral intelligence” are largely preserved, resulting in the apparent preservation of intellectual abilities described by many aphasia students. To support this claim, he reports some clinical vignettes, among them the famous “Pierre Maries’s cook” story. The “cuisinier”, who is affected by an aphasia of intermediate severity, is asked to fry some eggs in the hospital kitchen. “He starts by breaking the egg, in a very clumsy way; he pours it in a dish, without taking any precaution to avoid to break the yolk; he puts the butter on the dish, over the egg, then adds salt and pepper, and puts everything in the oven”¹.

Using contemporary terminology, Marie is proposing that aphasia is a disorder of semantic memory, including, in the cook case, conceptual knowledge about action sequences and tool use (ideational apraxia in naturalistic settings), a condition often observed in neurodegenerative disorders, in particular in the semantic variant of primary progressive aphasia, actually originally defined as “semantic dementia” (Hodges & Patterson 2007). These manifestations are however extremely rare in aphasia due to stroke (Ochipa,

1. The tone of the description, even considering the period, is unpleasantly patronizing. There is a curious story in the French Wikipedia article about Pierre Marie. After his retirement, 1928, he is reported to become engaged as a journalist in the socialist press, and an ardent advocate of physical activity for the masses. He also apparently authored a book “Pour le Sport Ouvrier”, and then mysteriously wrote some articles for a collaborationist paper after his death...

Rothi & Heilman 1992, Buxbaum 1998).

The status of the intellectual abilities of aphasia patients can be considered as one aspect of the classic debate on the relationship between language and thought (Carruthers & Boucher 1998). Neurological and neuropsychological research has addressed the issue through the administration of 'non-verbal' intelligence tests to aphasics. The "non-verbal" tests, however, necessarily involve the ability to understand verbal instructions, and furthermore generally contain material that can be verbalized. For example, if geometric shapes are presented to be memorised, a non-aphasic subject may have the advantage of being able to label their shape and their name, whereas an aphasic may not have access to this additional information. The results are not surprisingly heterogeneous, considering the multiple variables affecting aphasic patients' performance. For example, many studies fail to take into account the level of education, occupation and achievements in one's field of activity, i.e. the main determinants of cognitive reserve, the crucial resilience factor to the consequences of brain damage (Stern et al. 2022). While some patients perform as control subjects, others are impaired. In general, the severity of aphasia predicts the level of performance in group studies based on non-verbal intelligence tests. Several careful single-cases studies support the presence of residual high-level cognitive abilities in patients with severe aphasia. The case of SA, a retired policeman, shows an interesting contrast between a severe language disorder and the preserved ability to drive a car, play chess and make financial plans (Varley & Siegal 2000). The patient could demonstrate the ability to infer the mental states of others (he had an intact 'theory of mind') when an adaptation of classic false belief tests was administered. For example, he was shown a book, inside of which a hole had been made where a necklace had been placed. When asked, with the book closed, what a third person would think the book contained, when given the choice between a necklace and the pages of a book, he correctly indicated the second choice. This behaviour indicates that the ability to reason about the mental contents of others is not dependent on linguistic abilities. The same patient also demonstrated preserved inductive abilities about the causes of events, a central com-

ponent of scientific reasoning. For example, the patient was shown figures representing an event (e.g., a house on fire), and asked to indicate possible causes by any means of communication (words, but also gestures and drawings). His answers demonstrated a correct ability to identify plausible causes, both for probable situations but also for unexpected situations (e.g. a bullet bouncing off the skin of a buffalo, an event which was attributed to the use of a toy gun, or to the buffalo being made of wood) (Varley 2002). A second patient, SO, a science teacher, also affected by a severe aphasia, was able to solve algebraic problems, such as judging equivalence and performing transformations of expressions in algebraic notation (Klessinger, Szczerbinski & Varley 2007). The ability to reorient oneself in space in an ecological-like context after a disorientation procedure was demonstrated in a group of five patients with severe aphasia (Bek et al. 2010). These clinical observations share a crucial feature: rather than using traditional intelligence tests, they were based on ad hoc experimental procedures, where the use of language was minimised with respect to other modes of communication, such as gesture and drawing, to allow aphasia patients access to the information they need to manifest their cognitive processing abilities through the (limited) residual communication channels available to them.

Finally, the Pierre Marie's position on the neurological foundation of aphasia can be considered as strictly localizationist: "given that aphasia is one, its localization must necessarily be one. And it is really so. The only cerebral territory whose lesion produces aphasia is the so-called Wernicke's territory (gyrus supramarginalis, "pli courbe", and foot of the two first temporal gyri)". His focus on the role of white matter is actually limited to the recognition that a lesion of the fibres underlying the Wernicke's territory, isolating the cortex, has the same effects as damage to the cortex itself, according to localist version of the mass effect theory. In his words, "the severity of aphasia is proportional to the extent of damage to the cortex and underlying white matter". The strong tenet that "it is a mistake to attempt to dissociate this territory in different subregions" in Marie's view leads also to the denial of somatotopic and retinotopic representations in motor and visual cortex. Marie's view of language neuro-

biology can hardly be considered as in line with current large-scale network models, which are actually very much indebted to the work of Marie's antagonists in the famous debate that took place in a special meeting of the Paris Neurological Society after the publication of his papers, requiring three sessions in June and July 1908 (Klippel 1908), i.e. Madame Klumpke Déjerine and her husband, Jules Déjerine.

Conclusion

What we can safely conclude is that Pierre Marie may have scored some points on the part destruens of his discussion, i.e. his anatomically grounded criticism of the exclusive role of the third frontal convolution in language production. This topic, extensively discussed in the second and third paper, and the main focus of the Paris debate, has been central in Andrea Moro's neurolinguistic research, leading to novel insights, such as its role in inner speech (Magrassi et al. 2015) and in universal "possible languages" acquisition (Musso et al. 2003). On the other hand, he was severely defective in the pars construens of his "revision of the aphasia question", by assigning a central role to Wernicke's region in "intellectual function", rather than in language-specific processing.

References

- Bek, J, M Blades, M Siegal & R Varley. 2010. Language and spatial reorientation: evidence from severe aphasia. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 36(3). 646.
- Buxbaum, LJ. 1998. Ideational apraxia and naturalistic action. *Cognitive neuropsychology* 15(6-8). 617–43.
- Carruthers, P & J. Boucher. 1998. *Language and thought: Interdisciplinary themes*. Cambridge University Press.
- Coutinho, L, P Caramelli & HAG Teive. 2021. Aphasia localization: was Pierre Marie right? *Brain* 144(12). 3547–9.
- Duffy, RJ. & BZ. Liles. 1979. A translation of Finkelnburg's (1870) lecture on aphasia as "asymbolia" with commentary. *Journal of Speech and Hearing Disorders* 44(2). 156–68.
- Finkelnburg, F. 1870. Asymbolie und Aphasie. *Berliner Klinische Wochenschrift* 7. 449–50.
- Hodges, JR & K. Patterson. 2007. Semantic dementia: a unique clinicopathological syndrome. *The Lancet Neurology* 6(11). 1004–14.
- Klessinger, N, M Szczerbinski & R Varley. 2007. Algebra in a man with severe aphasia. *Neuropsychologia* 45(8). 1642–8.
- Klippel, M. 1908. Discussion sur l'aphasie. *Revue neurologique* 16(61). 1–636.
- Magrassi, L, G Aromataris, A Cabrini, V Annovazzi-Lodi & A Moro. 2015. Sound representation in higher language areas during language generation. *Proc Natl Acad Sci USA* 112(6). 1868–73.
- Marie, P. 1906a. L'aphasie de 1861-1866; essai de critique historique sur la genèse de la doctrine de Broca. *Semaine médicale* 26. 565–74.
- Marie, P. 1906b. La troisieme circonvolution frontale gauche ne joue aucun role special dans la fonction de langage. *La semaine médicale* 26. 241–7.
- Marie, P. 1906c. Que faut-il penser des aphasies sous-corticales (aphasies pures)? *La semaine médicale* 26. 493–500.

- Musso, M, A Moro, V Glauche, M Rijntjes, J Reichenbach, C. Büchel & C. Weiller. 2003. Broca's area and the language instinct. *Nat Neurosci* 6(7). 774–81.
- Ochipa, C, LJG Rothi & KM Heilman. 1992. Conceptual apraxia in Alzheimer's disease. *Brain* 115(4). 1061–71.
- Stern, Y, M Albert, C Barnes, R Cabeza, A Pascual-Leone & P. Rapp. 2022. A Framework for Concepts of Reserve and Resilience in Aging. *Neurobiology of Aging*.
- Varley, R. 2002. *Science without grammar: Scientific reasoning in severe agrammatic aphasia*.
- Varley, R & M. Siegal. 2000. Evidence for cognition without grammar from causal reasoning and 'theory of mind' in an agrammatic aphasic patient. *Current Biology* 10(12). 723–6.
- Weiller, C, V Glauche & M. Rijntjes. 2022. The ventral pathway and the extreme capsule: Pierre Marie was right. *Brain*.